Poster No. P08

A CROSS-SECTIONAL STUDY OF ASSESSING KNOWLEDGE, ATTITUDE, AND PRACTICE OF 5-LEVEL MALAYSIAN TRIAGE SCALE 2022 AMONG EMERGENCY HEALTH CARE WORKERS IN MALAYSIA

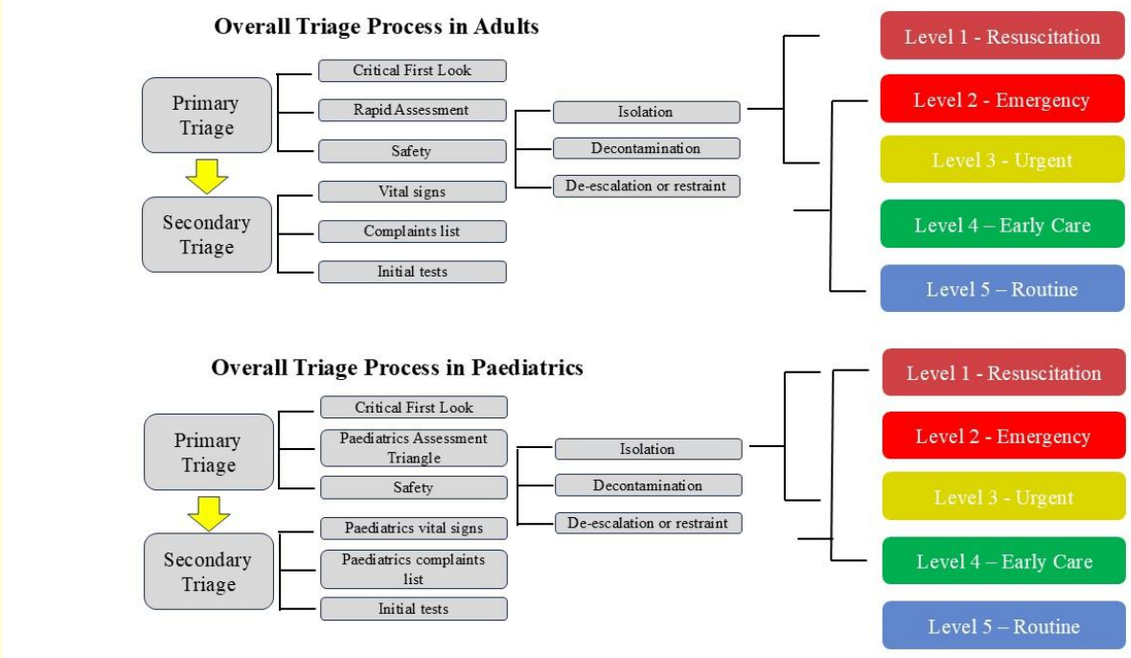
Muhammad Khidir Mohd Kamil[a], Rashidi Ahmad[b], Mohd Anizan Aziz[c], Samsu Ambia Ismail[c], Farah Alwi[d], Tirumal Subramaniam[e], Izruf Qabil[f]

[a]Emergency and Trauma Unit, Seri Iskandar Hospital, [b]Emergency and Trauma Department, KPJ Seremban Specialist Hospital, [c]Emergency and Trauma Department, Teluk Intan Hospital, [d]Department of Emergency Medicine and Trauma, Seri Manjung Hospital, [e]Department of Emergency Medicine and Trauma, Batu Gajah Hospital, [f]Department of Emergency Medicine and Trauma, Taiping Hospital

INTRODUCTION

The traditional framework of ED triage, which used a three-level (3L) triage system, was replaced with a five-level (5L) triage system due to a lack of agreement and inter-rater reliability. Historically, the MOH hospitals implemented the 3L color-coded Malaysian Triage Category (MTC). In 2022, the MOH issued updated guidelines for ED nationwide, promoting the 5L Malaysian Triage Scale (MTS). This study aims to evaluate the knowledge, attitude, and practice (KAP) regarding the MTS 2022 protocol among emergency health care workers (HCW).

MALAYSIAN TRIAGE SCALE 2022



MATERIALS & METHODS

- This prospective, cross-sectional study was conducted via a validated online questionnaire from Dec 1, 2023, to Nov 30, 2024.

Inclusion criteria included emergency physicians (EP), medical officers (MO), nurses, and assistant medical officers (AMO) across ED in the MOH hospitals.

Incomplete forms and responses were excluded

- Using convenience sampling, a Google Forms link was distributed digitally via email and WhatsApp.
- The sample size was estimated using Krejcie (1970) method - calculated sample size was 370.
- The questionnaire was validated by five external reviewers - content validity indexes of 0.90, 0.85, and 0.85 for the KAP domains, respectively, with an overall Cronbach's alpha of 0.80. The reliability of the test and re-test using intraclass correlation coefficient, yielding a value of 0.90.
- The instrument comprised 43 items: knowledge and practice domains were scored binary (1 for correct/standard, 0 for incorrect/substandard), while the attitude domain utilized a 5-point Likert scale.
- The analysis of the results was conducted utilizing SPSS V. 26. Knowledge and practice levels were categorized using Bloom's cut-off points: good (>80%), average (60%–79.9%), and poor (<60%). Attitude scores were classified as positive, neutral, or negative. Non-parametric continuous data were analysed using Mann-Whitney U or Kruskal-Wallis tests, with subsequent post-hoc analysis employing Bonferroni correction. The correlations between each of the KAP domains, were examined using Spearman's rank correlation
- This study obtained ethical approval from the MOH MREC (NMRR ID-23-03110-UPE).

QUESTIONNAIRE

RESULTS

DEMOGRAPHICS

370 included
37 excluded

50.5% female
32 (21-55) median age

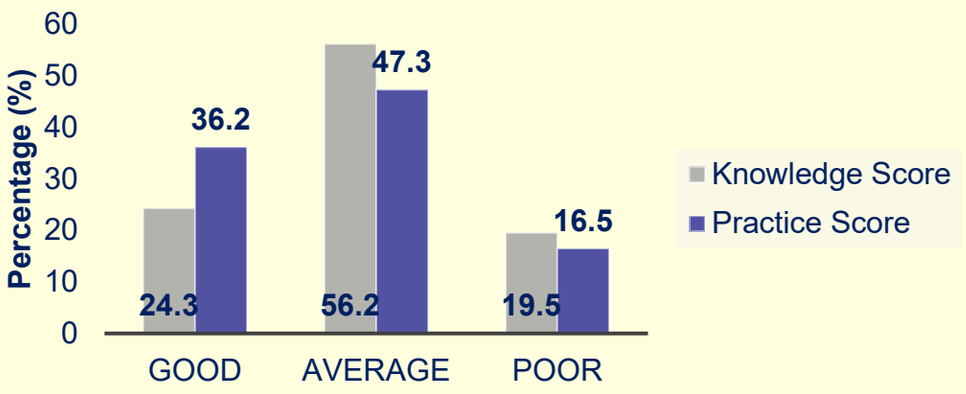
38.1% AMO, 18.9% nurse, 37.0 MO, & 5.9% EP

52.4% diploma, 39.2% bachelor, 8.1% masters degree, & 0.3% doctorate

31.3% type 4, 24.3% type 3, 25.7% type 2, & 18.9% type 1 hospitals

OUTCOMES

The levels of knowledge & practice scores



60.5% confident to perform MTS
77.9% motivated to learn MTS
54.6% competent to perform MTS
70.6% confident to perform critical first look & rapid assessment
62.1% confident to perform PAT

- Average performance in executing triaging for level 1 / 2 (60%), whereas only 20% respondents were capable to accurately identify patients categorized as level 4 & 5.
- Post-hoc Bonferroni analysis showed median knowledge scores differed significantly by designation (p=0.001) and education (p= 0.001).
- Weak positive correlations were found between knowledge and attitude (r_s=0.165, p= 0.001) and knowledge and practice (r_s=0.319, p=0.001).

DISCUSSION / CONCLUSION

Emergency HCWs exhibit sufficient knowledge and practice levels, alongside overall favourable attitudes toward the MTS 2022 protocol. While motivation is high, critical gaps remain in classification proficiency for low-acuity cases and individual competence confidence. Targeted educational interventions through simulation-based training are recommended, tailored to different ED staff designations and educational backgrounds, to bridge knowledge gaps and ensure successful nationwide triage standardization.

- We thank the external reviewers for their validation of this study.
- All authors declare no conflicts of interests.

[1] Travers, et al. Five-level triage system more effective than three-level in tertiary emergency department. J Emerg Nurs 2002;**28**(5):395-400.
[2] Emergency Medicine and Trauma Services MoH, Malaysia. Malaysian Triage Scale 2022 for Emergency and Trauma Departments. 2023.